



Experiment 04: Software Provisioning and Configuration using Ansible

Experiment No.	04
Mapped CO	CO4
Aim	To automate the provisioning and configuration of software on target hosts using Ansible.
Tools / Software	Ansible; One or more Linux hosts / VMs reachable over SSH; SSH key pair; Terminal / editor

Theory

Ansible is an agentless configuration-management and automation tool used to provision, configure and manage servers. It connects to managed nodes mainly through SSH and applies the desired system state written in YAML files. Since no agent is required on the target host, Ansible is simple to start with and suitable for repeatable software configuration tasks.

The main components of Ansible are inventory, modules, playbooks and roles. An inventory lists the target hosts, modules perform specific tasks such as installing a package or starting a service, and playbooks describe ordered tasks to be executed on selected hosts. Idempotence is an important property of Ansible: running the same playbook multiple times should not repeatedly change a system that is already in the desired state.

Procedure

1. Install Ansible on the control node and verify the installation using `ansible --version`.
2. Create or identify one Linux managed host / VM that is reachable from the control node using SSH.
3. Configure passwordless SSH access using an SSH key pair or ensure that valid SSH credentials are available.
4. Create an inventory file named `inventory.ini` and list the target host under a group such as `[web]`.
5. Verify connectivity to the managed host using the Ansible ping module.
6. Write a playbook named `site.yml` to install and start a web server such as `nginx` and deploy a simple web page.
7. Run the playbook using `ansible-playbook` and observe the ok / changed / failed task summary.
8. Re-run the playbook to verify idempotence and confirm that no unnecessary changes are made.

Sample Code / Commands

inventory.ini

```
[web]
192.168.56.10 ansible_user=ubuntu
```

site.yml

```
- name: Configure web servers
  hosts: web
  become: true
  tasks:
    - name: Install nginx
      apt:
        name: nginx
        state: present
```



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```
update_cache: true

- name: Ensure nginx is running
  service:
    name: nginx
    state: started
    enabled: true

- name: Deploy index page
  copy:
    content: "<h1>Welcome to Ansible Lab</h1>"
    dest: /var/www/html/index.html
```

Run commands

```
ansible --version
ansible -i inventory.ini web -m ping
ansible-playbook -i inventory.ini site.yml
ansible-playbook -i inventory.ini site.yml
```



Observation / Experiment Output

Instructions to students: Paste screenshots or copied outputs in the boxes below after completing Ansible inventory creation, connectivity testing, playbook execution and verification. Keep screenshots clear and readable.

Name	Roll No.	Batch	Date
Vighnesh Pingale	48	B2	14/08/26

1. Ansible Installation and Version Output

```
en_PH.utf8
en_SG.utf8
en_US.utf8
en_ZA.utf8
en_ZM.utf8
en_ZW.utf8
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ export LANG=en_IN.UTF-8
export LANGUAGE=en_IN:en
export LC_ALL=en_IN.UTF-8
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ locale
LANG=en_IN.UTF-8
LANGUAGE=en_IN:en
LC_CTYPE="en_IN.UTF-8"
LC_NUMERIC="en_IN.UTF-8"
LC_TIME="en_IN.UTF-8"
LC_COLLATE="en_IN.UTF-8"
LC_MONETARY="en_IN.UTF-8"
LC_MESSAGES="en_IN.UTF-8"
LC_PAPER="en_IN.UTF-8"
LC_NAME="en_IN.UTF-8"
LC_ADDRESS="en_IN.UTF-8"
LC_TELEPHONE="en_IN.UTF-8"
LC_MEASUREMENT="en_IN.UTF-8"
LC_IDENTIFICATION="en_IN.UTF-8"
LC_ALL=en_IN.UTF-8
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ ansible --version
ansible [core 2.14.18]
  config file = None
  configured module search path = ['/home/apsit/.ansible/plugins/modules', '/usr/share/
  ansible/plugins/modules']
  ansible python module location = /home/apsit/.local/pipx/venvs/ansible-core/lib/pytho
  n3.10/site-packages/ansible
  ansible collection location = /home/apsit/.ansible/collections:/usr/share/ansible/col
  lections
  executable location = /home/apsit/.local/bin/ansible
  python version = 3.10.12 (main, Jun 22 2026, 18:55:27) [GCC 11.4.0] (/home/apsit/.loc
  al/pipx/venvs/ansible-core/bin/python)
  jinja version = 3.1.6
  libyaml = True
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ |
```

```
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC: ~/ansible-nginx-de...
apsit@apsit-HP-Pro-Towe... x  apsit@apsit-HP-Pro-Towe... x  apsit@apsit-HP-Pro-Tower... x
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-136-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

1 device has a firmware upgrade available.
Run 'fwupdmgrr get-upgrades' for more information.

Expanded Security Maintenance for Applications is not enabled.

6 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

32 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

1 device has a firmware upgrade available.
Run 'fwupdmgrr get-upgrades' for more information.

Last login: Fri Aug 14 08:23:49 2026 from 192.168.75.12
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ hostname
apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ exit
logout
Connection to 192.168.75.12 closed.
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ mkdir -p ~/ansible-nginx-demo
cd ~/ansible-nginx-demo
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ pwd
/home/apsit/ansible-nginx-demo
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ nano inventory.ini
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ cat inventory.ini
[web]
192.168.75.12 ansible_user=apsit
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ |
```



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2. Inventory File Output

```
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC: ~/ansible-nginx-de...
apsit@apsit-HP-Pro-Towe... x  apsit@apsit-HP-Pro-Towe... x  apsit@apsit-HP-Pro-Tower... x
LANG=en_IN.UTF-8
LANGUAGE=en_IN:en
LC_CTYPE="en_IN.UTF-8"
LC_NUMERIC="en_IN.UTF-8"
LC_TIME="en_IN.UTF-8"
LC_COLLATE="en_IN.UTF-8"
LC_MONETARY="en_IN.UTF-8"
LC_MESSAGES="en_IN.UTF-8"
LC_PAPER="en_IN.UTF-8"
LC_NAME="en_IN.UTF-8"
LC_ADDRESS="en_IN.UTF-8"
LC_TELEPHONE="en_IN.UTF-8"
LC_MEASUREMENT="en_IN.UTF-8"
LC_IDENTIFICATION="en_IN.UTF-8"
LC_ALL=en_IN.UTF-8
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ ansible --version
ansible [core 2.14.18]
  config file = None
  configured module search path = ['/home/apsit/.ansible/plugins/modules', '/usr/share/
  ansible/plugins/modules']
  ansible python module location = /home/apsit/.local/pipx/venvs/ansible-core/lib/pytho
  n3.10/site-packages/ansible
  ansible collection location = /home/apsit/.ansible/collections:/usr/share/ansible/col
  lections
  executable location = /home/apsit/.local/bin/ansible
  python version = 3.10.12 (main, Jun 22 2026, 18:55:27) [GCC 11.4.0] (/home/apsit/.loc
  al/pipx/venvs/ansible-core/bin/python)
  jinja version = 3.1.6
  libyaml = True
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ cd ~/ansible-nginx
-demo
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ ansible -i inventory.ini web -m ping
192.168.75.12 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ |
```

```
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC: ~/ansible-nginx-de...
apsit@apsit-HP-Pro-Towe... x  apsit@apsit-HP-Pro-Towe... x  apsit@apsit-HP-Pro-Tower... x
lections
executable location = /home/apsit/.local/bin/ansible
python version = 3.10.12 (main, Jun 22 2026, 18:55:27) [GCC 11.4.0] (/home/apsit/.loc
al/pipx/venvs/ansible-core/bin/python)
jinja version = 3.1.6
libyaml = True
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ cd ~/ansible-nginx
-demo
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ ansible -i inventory.ini web -m ping
192.168.75.12 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ nano site.yml
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ cat site.yml
- name: Configure web servers
  hosts: web
  become: true

  tasks:
    - name: Install nginx
      apt:
        name: nginx
        state: present
        update_cache: true

    - name: Ensure nginx is running
      service:
        name: nginx
        state: started
        enabled: true

    - name: Deploy index page
      copy:
        content: "<h1>Welcome to Ansible Lab</h1>"
        dest: /var/www/html/index.html
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ |
```




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3. SSH Connectivity / Ping Module Output

```
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC: ~  
apsit@apsit-HP-Pro-Towe... x  apsit@apsit-HP-Pro-Towe... x  apsit@apsit-HP-Pro-Tower... x  
Hint: Some lines were ellipsized, use -l to show in full.  
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ ssh apsit@192.168.75.12  
The authenticity of host '192.168.75.12 (192.168.75.12)' can't be established.  
ED25519 key fingerprint is SHA256:f24s7Rlj9dC1qXHoLpuewctONQtdWhGGUCEDnxmpXkc.  
This host key is known by the following other names/addresses:  
~/.ssh/known_hosts:1: [hashed name]  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '192.168.75.12' (ED25519) to the list of known hosts.  
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-136-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
1 device has a firmware upgrade available.  
Run 'fwupdmgr get-upgrades' for more information.  
  
Expanded Security Maintenance for Applications is not enabled.  
  
6 updates can be applied immediately.  
To see these additional updates run: apt list --upgradable  
  
32 additional security updates can be applied with ESM Apps.  
Learn more about enabling ESM Apps service at https://ubuntu.com/esm  
  
New release '24.04.4 LTS' available.  
Run 'do-release-upgrade' to upgrade to it.  
  
1 device has a firmware upgrade available.  
Run 'fwupdmgr get-upgrades' for more information.  
  
Last login: Thu Aug 13 11:00:45 2026 from 192.168.21.187  
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ hostname  
apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC  
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ exit  
logout  
Connection to 192.168.75.12 closed.  
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ |
```

```
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC: ~  
apsit@apsit-HP-Pro-Towe... x  apsit@apsit-HP-Pro-Towe... x  apsit@apsit-HP-Pro-Tower... x  
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ hostname -I  
192.168.75.12 192.168.25.121 172.17.0.1  
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ whoami  
apsit  
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ hostname  
apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC  
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ sudo apt install openssh-server -y  
[sudo] password for apsit:  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
openssh-server is already the newest version (1:8.9p1-3ubuntu0.16).  
The following packages were automatically installed and are no longer required:  
  libfwupd2 libfwupdplugins libgcb-1.0-0 libmbios-c2  
Use 'sudo apt autoremove' to remove them.  
0 upgraded, 0 newly installed, 0 to remove and 6 not upgraded.  
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ sudo systemctl enable --now ssh  
Synchronizing state of ssh.service with SysV service script with /lib/systemd/systemd-s  
ysv-install.  
Executing: /lib/systemd/systemd-sysv-install enable ssh  
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ systemctl status ssh --no-pager  
● ssh.service - OpenBSD Secure Shell server  
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)  
   Active: active (running) since Fri 2026-08-14 08:07:06 IST; 15min ago  
     Docs: man:sshd(8)  
           man:sshd_config(5)  
    Main PID: 1187 (sshd)  
      Tasks: 1 (limit: 8948)  
    Memory: 2.8M  
       CPU: 19ms  
    CGroup: /system.slice/ssh.service  
            └─1187 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"  
  
Aug 14 08:07:06 apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC systemd[1]: Starting OpenBS...  
Aug 14 08:07:06 apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC sshd[1187]: Server listenin...2.  
Aug 14 08:07:06 apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC sshd[1187]: Server listenin...2.  
Aug 14 08:07:06 apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC systemd[1]: Started OpenBSD...r.  
Hint: Some lines were ellipsized, use -l to show in full.  
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ |
```



4. Playbook File Output

```
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC: ~/ansible-nginx-de...  
apsit@apsit-HP-Pro-Towe... x  apsit@apsit-HP-Pro-Towe... x  apsit@apsit-HP-Pro-Tower... x  v  
192.168.75.12 : ok=0  changed=0  unreachable=0  failed=1  skipped=  
0  rescued=0  ignored=0  
  
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ ansible-playbook -  
i inventory.ini site.yml --ask-become-pass  
BECOME password:  
  
PLAY [Configure web servers] *****  
  
TASK [Gathering Facts] *****  
ok: [192.168.75.12]  
  
TASK [Install nginx] *****  
changed: [192.168.75.12]  
  
TASK [Ensure nginx is running] *****  
ok: [192.168.75.12]  
  
TASK [Deploy index page] *****  
changed: [192.168.75.12]  
  
PLAY RECAP *****  
192.168.75.12 : ok=4  changed=2  unreachable=0  failed=0  skipped=  
0  rescued=0  ignored=0  
  
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ curl http://192.16  
8.75.12  
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ curl -I http://192  
.168.75.12o$ curl -I http://192.168.75.12  
HTTP/1.1 200 OK  
Server: nginx/1.18.0 (Ubuntu)  
Date: Fri, 14 Aug 2026 03:04:07 GMT  
Content-Type: text/html  
Content-Length: 31  
Last-Modified: Fri, 14 Aug 2026 03:03:21 GMT  
Connection: keep-alive  
ETag: "6a7e8579-1f"  
Accept-Ranges: bytes  
  
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ |
```



5. Playbook Execution and Idempotence Output

```
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC: ~/ansible-nginx-de...  
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ curl http://192.168.75.12  
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ curl -I http://192.168.75.12  
HTTP/1.1 200 OK  
Server: nginx/1.18.0 (Ubuntu)  
Date: Fri, 14 Aug 2026 03:04:07 GMT  
Content-Type: text/html  
Content-Length: 31  
Last-Modified: Fri, 14 Aug 2026 03:03:21 GMT  
Connection: keep-alive  
ETag: "6a7e8579-1f"  
Accept-Ranges: bytes  
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ curl -s http://192.168.75.12 | cat  
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ ansible-playbook -i inventory.ini site.yml --ask-become-pass  
BECOME password:  
PLAY [Configure web servers] *****  
TASK [Gathering Facts] *****  
ok: [192.168.75.12]  
TASK [Install nginx] *****  
ok: [192.168.75.12]  
TASK [Ensure nginx is running] *****  
ok: [192.168.75.12]  
TASK [Deploy index page] *****  
ok: [192.168.75.12]  
PLAY RECAP *****  
192.168.75.12 : ok=4 changed=0 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0  
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ |
```

```
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC: ~/ansible-nginx-de...  
ok: [192.168.75.12]  
TASK [Install nginx] *****  
ok: [192.168.75.12]  
TASK [Ensure nginx is running] *****  
ok: [192.168.75.12]  
TASK [Deploy index page] *****  
ok: [192.168.75.12]  
PLAY RECAP *****  
192.168.75.12 : ok=4 changed=0 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0  
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ cat /var/www/html/index.html  
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ curl -v http://192.168.75.12  
* Trying 192.168.75.12:80...  
* Connected to 192.168.75.12 (192.168.75.12) port 80 (#0)  
> GET / HTTP/1.1  
> Host: 192.168.75.12  
> User-Agent: curl/7.81.0  
> Accept: */*  
>  
* Mark bundle as not supporting multiuse  
< HTTP/1.1 200 OK  
< Server: nginx/1.18.0 (Ubuntu)  
< Date: Fri, 14 Aug 2026 03:07:03 GMT  
< Content-Type: text/html  
< Content-Length: 31  
< Last-Modified: Fri, 14 Aug 2026 03:03:21 GMT  
< Connection: keep-alive  
< ETag: "6a7e8579-1f"  
< Accept-Ranges: bytes  
<  
* Connection #0 to host 192.168.75.12 left intact  
<h1>Welcome to Ansible Lab</h1>  
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ |
```



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GitHub Repository / Lab Folder Link:

Local Lab Folder: /home/apsit/ansible-nginx-demo

Managed Host / VM Details:

Managed Host IP: 192.168.75.12

Username: apsit

Operating System: Ubuntu 22.04.5 LTS

Hostname: apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC

SSH: Passwordless SSH configured

Web Server: nginx

Final Observation Statement:

Ansible was successfully used to connect to the managed Ubuntu host through SSH and automate nginx installation, service configuration, and web-page deployment. The nginx page was successfully verified using curl with HTTP 200 OK. On the second playbook execution, changed=0 was observed, demonstrating idempotent configuration management.

Expected Output

The managed host should have nginx installed and running. The web server should serve the deployed index page, the Ansible playbook should complete successfully, and the second run should show idempotent behaviour with no unnecessary changes.

Conclusion

Software provisioning and configuration were automated declaratively with Ansible, demonstrating agentless management, repeatable configuration and idempotent convergence to the desired system state.

Viva / Review Questions

1. Why is Ansible described as agentless?
2. What does idempotence mean in configuration management?
3. Differentiate a playbook from a role.
4. What transport does Ansible use to reach Linux managed nodes?

Marks: ____ / 10

Date: _____ **Signature of Faculty:** _____